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Experiment 03

Aim: To deploy multi-service applications using Docker Compose.

LO4: To deploy and manage containers using Kubernetes.

Theory:

What is Docker Compose?
Docker Compose is a tool used to define and manage multi-container Docker applications using a single YAML file (docker-compose.yml). Instead of creating and managing each container separately with individual Docker commands, Docker Compose allows all services to be configured in one file and started with a single command.
It simplifies application deployment by automatically creating the required containers, networks, and volumes, ensuring that all services work together seamlessly.

Why Use Docker Compose?

- Simplifies the deployment of multi-container applications.
- Allows all services to be managed from a single configuration file.
- Automatically creates a dedicated network for communication between containers.
- Supports persistent storage using Docker volumes.
- Makes applications portable and easy to reproduce across different systems.

Services
A service represents a container or a group of identical containers defined in the Compose file.
Example:

```
services:
  web:
  database:
```

Here,

- web creates an Nginx container.
- database creates a MySQL container.

- Easy deployment of multi-container applications.

Command	Explanation
docker compose config	Validates and displays the configuration of the docker-compose.yml file.
docker compose up	Creates and starts all services defined in the Compose file.
docker compose up -d	Creates and starts all services in detached (background) mode.
docker compose ps	Displays the status of all containers in the Compose project.
docker compose logs	Displays the log output of the running containers.
docker compose stop	Stops all running containers without removing them.
docker compose start	Starts previously stopped containers.
docker compose restart	Restarts all running containers.
docker compose down	Stops and removes all containers, networks, and other resources created by Docker Compose.
docker compose down -v	Stops and removes all containers along with their associated Docker volumes.
docker compose images	Lists the Docker images used by the Compose project.

Commands:

docker –version

```
C:\Users\15L>docker --version
Docker version 25.0.3, build 4debf41

C:\Users\15L>docker compose version
Docker Compose version v2.24.5-desktop.1
```

Create a Folder
mkdir compose-demo

Go Inside the Folder
cd compose-demo

```
C:\Users\15L>mkdir compose-demo

C:\Users\15L>cd compose-demo

C:\Users\15L\compose-demo>notepad
```

Paste the Docker Compose File

```
services:
  web:
    image: nginx
    ports:
      - "8081:80"

  database:
    image: mysql:8.0
    environment:
      MYSQL_ROOT_PASSWORD: root123

  redis:
    image: redis

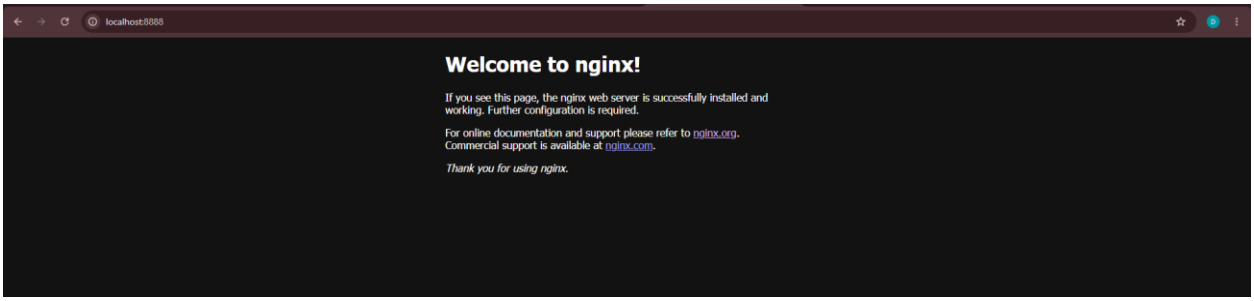
  phpmyadmin:
    image: phpmyadmin
    ports:
      - "8082:80"
    environment:
      PMA_HOST: database
```

docker compose down

```
C:\Users\15L\compose-demo>docker compose down
[+] Running 5/5
 ✔Container compose-demo-redis-1       Removed
 ✔Container compose-demo-web-1         Removed
 ✔Container compose-demo-database-1    Removed
 ✔Container compose-demo-phpmyadmin-1  Removed
 ✔Network compose-demo_default         Removed
```

docker compose up -d

```
C:\Users\15L\compose-demo>docker compose up -d
[+] Running 4/5
 - Network compose-demo_default           Created
 ✔Container compose-demo-database-1       Started
 ✔Container compose-demo-web-1            Started
 ✔Container compose-demo-phpmyadmin-1     Started
 ✔Container compose-demo-redis-1          Started
```



docker compose config

```
C:\Users\15L\compose-demo>docker compose config
name: compose-demo
services:
  database:
    environment:
      MYSQL_ROOT_PASSWORD: root123
    image: mysql:8.0
    networks:
      default: null
  phpmyadmin:
    environment:
      PMA_HOST: database
    image: phpmyadmin
    networks:
      default: null
    ports:
      - mode: ingress
        target: 80
        published: "8082"
        protocol: tcp
  redis:
    image: redis
    networks:
      default: null
  web:
    image: nginx
    networks:
      default: null
    ports:
      - mode: ingress
        target: 80
        published: "8888"
        protocol: tcp
networks:
  default:
    name: compose-demo_default
```

docker compose up -d

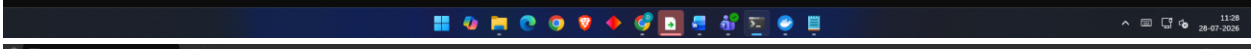
```
C:\Users\15L\compose-demo>docker compose up -d
[+] Running 4/0
 ✔Container compose-demo-phpmyadmin-1  Running
 ✔Container compose-demo-web-1         Running
 ✔Container compose-demo-redis-1       Running
 ✔Container compose-demo-database-1    Running
```

docker compose ps

```
C:\Users\15L\compose-demo>docker compose ps
NAME                      IMAGE        COMMAND                  SERVICE      CREATED          STATUS          PORTS
compose-demo-database-1   mysql:8.0    "docker-entrypoint.s…"   database     24 seconds ago   Up 23 seconds   3306/tcp, 33060/tcp
compose-demo-phpmyadmin-1 phpmyadmin   "/docker-entrypoint.…"   phpmyadmin   24 seconds ago   Up 23 seconds   0.0.0.0:8082->80/tcp
compose-demo-redis-1      redis        "docker-entrypoint.s…"   redis        24 seconds ago   Up 23 seconds   6379/tcp
compose-demo-web-1        nginx        "/docker-entrypoint.…"   web          24 seconds ago   Up 23 seconds   0.0.0.0:8888->80/tcp
```

docker compose logs

```
C:\Users\15L\compose-demo>docker compose logs
web-1  | /docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
web-1  | /docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
web-1  | /docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
web-1  | 10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
web-1  | 10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
web-1  | /docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
web-1  | /docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
web-1  | /docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
web-1  | /docker-entrypoint.sh: Configuration complete; ready for start up
web-1  | 2026/07/28 05:53:34 [notice] 1#1: using the "epoll" event method
web-1  | 2026/07/28 05:53:34 [notice] 1#1: nginx/1.29.5
web-1  | 2026/07/28 05:53:34 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
web-1  | 2026/07/28 05:53:34 [notice] 1#1: OS: Linux 6.18.33.2-microsoft-standard-WSL2
web-1  | 2026/07/28 05:53:34 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
web-1  | 2026/07/28 05:53:34 [notice] 1#1: start worker processes
web-1  | 2026/07/28 05:53:34 [notice] 1#1: start worker process 29
web-1  | 2026/07/28 05:53:34 [notice] 1#1: start worker process 30
web-1  | 2026/07/28 05:53:34 [notice] 1#1: start worker process 31
web-1  | 2026/07/28 05:53:34 [notice] 1#1: start worker process 32
web-1  | 2026/07/28 05:53:34 [notice] 1#1: start worker process 33
web-1  | 2026/07/28 05:53:34 [notice] 1#1: start worker process 34
web-1  | 2026/07/28 05:53:34 [notice] 1#1: start worker process 35
web-1  | 2026/07/28 05:53:34 [notice] 1#1: start worker process 36
web-1  | 2026/07/28 05:53:34 [notice] 1#1: start worker process 37
web-1  | 2026/07/28 05:53:34 [notice] 1#1: start worker process 38
web-1  | 2026/07/28 05:53:34 [notice] 1#1: start worker process 39
web-1  | 2026/07/28 05:53:34 [notice] 1#1: start worker process 40
web-1  | 2026/07/28 05:53:34 [notice] 1#1: start worker process 41
web-1  | 2026/07/28 05:53:34 [notice] 1#1: start worker process 42
web-1  | 2026/07/28 05:53:34 [notice] 1#1: start worker process 43
web-1  | 2026/07/28 05:53:34 [notice] 1#1: start worker process 44
phpmyadmin-1 | AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 172.25.0.4. Set the 'ServerName' directive globally to suppress this message
redis-1      | Starting Redis Server
redis-1      | 1:C 28 Jul 2026 05:53:33.938 * oO0OoO0OoO0Oo Redis is starting oO0OoO0OoO0Oo
web-1  | 2026/07/28 05:53:34 [notice] 1#1: start worker process 45
phpmyadmin-1 | AH00558: apache2: Could not reliably determine the server's fully qualified domain name, using 172.25.0.4. Set the 'ServerName' directive globally to suppress this message
database-1   | 2026-07-28 05:53:34+00:00 [Note] [Entrypoint]: Entrypoint script for MySQL Server 8.0.46-1.el9 started.
redis-1      | 1:C 28 Jul 2026 05:53:33.938 * Redis version=8.8.1, bits=64, commit=00000000, modified=1, pid=1, just started
redis-1      | 1:C 28 Jul 2026 05:53:33.938 * Configuration loaded
redis-1      | 1:M 28 Jul 2026 05:53:33.938 * monotonic clock: POSIX clock_gettime
redis-1      | 1:M 28 Jul 2026 05:53:33.939 * Running mode=standalone, port=6379.
redis-1      | 1:M 28 Jul 2026 05:53:33.940 * <bf> RedisBloom version 8.8.2 (Git=unknown)
redis-1      | 1:M 28 Jul 2026 05:53:33.940 * <bf> Registering configuration options: [
redis-1      | 1:M 28 Jul 2026 05:53:33.940 * <bf>     { bf-error-rate       :      0.01 }
```



```
Administrator: Command Pro
database-1   | 2026-07-28 05:53:34+00:00 [Note] [Entrypoint]: Initializing database files
database-1   | 2026-07-28T05:53:34.292831Z 0 [Warning] [MY-011068] [Server] The syntax '--skip-host-cache' is deprecated and will be removed in a future release. Please use SET GLOBAL host_cache_size=0 instead.
database-1   | 2026-07-28T05:53:34.292885Z 0 [System] [MY-013169] [Server] /usr/sbin/mysqld (mysqld 8.0.46) initializing of server in progress as process 80
database-1   | 2026-07-28T05:53:34.297340Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.
database-1   | 2026-07-28T05:53:34.544758Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
database-1   | 2026-07-28T05:53:35.300509Z 6 [Warning] [MY-010453] [Server] root@localhost is created with an empty password ! Please consider switching off the --initialize-insecure option.
database-1   | 2026-07-28 05:53:37+00:00 [Note] [Entrypoint]: Database files initialized
database-1   | 2026-07-28 05:53:37+00:00 [Note] [Entrypoint]: Starting temporary server
database-1   | 2026-07-28T05:53:37.902478Z 0 [Warning] [MY-011068] [Server] The syntax '--skip-host-cache' is deprecated and will be removed in a future release. Please use SET GLOBAL host_cache_size=0 instead.
database-1   | 2026-07-28T05:53:37.903374Z 0 [System] [MY-010116] [Server] /usr/sbin/mysqld (mysqld 8.0.46) starting as process 124
database-1   | 2026-07-28T05:53:37.925546Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.
database-1   | 2026-07-28T05:53:38.049432Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
database-1   | 2026-07-28T05:53:38.175472Z 0 [Warning] [MY-010068] [Server] CA certificate ca.pem is self signed.
database-1   | 2026-07-28T05:53:38.175513Z 0 [System] [MY-013602] [Server] Channel mysql_main configured to support TLS. Encrypted connections are now supported for this channel.
database-1   | 2026-07-28T05:53:38.177189Z 0 [Warning] [MY-011810] [Server] Insecure configuration for --pid-file: Location '/var/run/mysqld' in the path is accessible to all OS users. Consider choosing a different directory.
database-1   | 2026-07-28T05:53:38.187578Z 0 [System] [MY-011323] [Server] X Plugin ready for connections. Socket: /var/run/mysqld/mysqlx.sock
database-1   | 2026-07-28T05:53:38.187690Z 0 [System] [MY-010931] [Server] /usr/sbin/mysqld: ready for connections. Version: '8.0.46'  socket: '/var/run/mysqld/mysqld.sock'  port: 0  MySQL Community Server - GPL.
database-1   | 2026-07-28 05:53:38+00:00 [Note] [Entrypoint]: Temporary server started.
database-1   | '/var/lib/mysql/mysql.sock' -> '/var/run/mysqld/mysqld.sock'
database-1   | Warning: Unable to load '/usr/share/zoneinfo/iso3166.tab' as time zone. Skipping it.
database-1   | Warning: Unable to load '/usr/share/zoneinfo/leap-seconds.list' as time zone. Skipping it.
database-1   | Warning: Unable to load '/usr/share/zoneinfo/leapseconds' as time zone. Skipping it.
database-1   | Warning: Unable to load '/usr/share/zoneinfo/tzdata.zi' as time zone. Skipping it.
database-1   | Warning: Unable to load '/usr/share/zoneinfo/zone.tab' as time zone. Skipping it.
database-1   | Warning: Unable to load '/usr/share/zoneinfo/zone1970.tab' as time zone. Skipping it.
database-1   |
database-1   | 2026-07-28 05:53:39+00:00 [Note] [Entrypoint]: Stopping temporary server
database-1   | 2026-07-28T05:53:39.493544Z 10 [System] [MY-013172] [Server] Received SHUTDOWN from user root. Shutting down mysqld (Version: 8.0.46).
database-1   | 2026-07-28T05:53:40.835368Z 0 [System] [MY-010910] [Server] /usr/sbin/mysqld: Shutdown complete (mysqld 8.0.46)  MySQL Community Server - GPL.
database-1   | 2026-07-28 05:53:41+00:00 [Note] [Entrypoint]: Temporary server stopped
database-1   |
database-1   | 2026-07-28 05:53:41+00:00 [Note] [Entrypoint]: MySQL init process done. Ready for start up.
database-1   |
database-1   | 2026-07-28T05:53:41.711333Z 0 [Warning] [MY-011068] [Server] The syntax '--skip-host-cache' is deprecated and will be removed in a future release. Please use SET GLOBAL host_cache_size=0 instead.
database-1   | 2026-07-28T05:53:41.712840Z 0 [System] [MY-010116] [Server] /usr/sbin/mysqld (mysqld 8.0.46) starting as process 1
database-1   | 2026-07-28T05:53:41.726729Z 1 [System] [MY-013576] [InnoDB] InnoDB initialization has started.
database-1   | 2026-07-28T05:53:41.851588Z 1 [System] [MY-013577] [InnoDB] InnoDB initialization has ended.
database-1   | 2026-07-28T05:53:41.942046Z 0 [Warning] [MY-010068] [Server] CA certificate ca.pem is self signed.
database-1   | 2026-07-28T05:53:41.942484Z 0 [System] [MY-013602] [Server] Channel mysql_main configured to support TLS. Encrypted connections are now supported for this channel.
database-1   | 2026-07-28T05:53:41.944360Z 0 [Warning] [MY-011810] [Server] Insecure configuration for --pid-file: Location '/var/run/mysqld' in the path is accessible to all OS users. Consider choosing a different directory.
database-1   | 2026-07-28T05:53:41.955762Z 0 [System] [MY-011323] [Server] X Plugin ready for connections. Bind-address: '::' port: 33060, socket: /var/run/mysqld/mysqlx.sock
database-1   | 2026-07-28T05:53:41.955777Z 0 [System] [MY-010931] [Server] /usr/sbin/mysqld: ready for connections. Version: '8.0.46'  socket: '/var/run/mysqld/mysqld.sock'  port: 3306  MySQL Community Server - GPL.
phpmyadmin-1 | [Tue Jul 28 05:53:34.172361 2026] [core:notice] [pid 1:tid 1] AH00094: Command line: 'apache2 -D FOREGROUND'
```

docker compose stop

```
C:\Users\15L\compose-demo>docker compose stop
[+] Stopping 4/4
 ✔Container compose-demo-database-1    Stopped
 ✔Container compose-demo-phpmyadmin-1  Stopped
 ✔Container compose-demo-web-1         Stopped
 ✔Container compose-demo-redis-1       Stopped
```

docker compose start

```
C:\Users\15L\compose-demo>docker compose start
[+] Running 4/4
 ✔Container compose-demo-database-1    Started
 ✔Container compose-demo-phpmyadmin-1  Started
 ✔Container compose-demo-redis-1       Started
 ✔Container compose-demo-web-1         Started
```

docker compose restart

```
C:\Users\15L\compose-demo>docker compose restart
[+] Restarting 4/4
 ✔Container compose-demo-database-1    Started
 ✔Container compose-demo-redis-1       Started
 ✔Container compose-demo-phpmyadmin-1  Started
 ✔Container compose-demo-web-1         Started
```

docker compose down

```
C:\Users\15L\compose-demo>docker compose down
[+] Running 5/5
 ✔Container compose-demo-redis-1       Removed
 ✔Container compose-demo-web-1         Removed
 ✔Container compose-demo-phpmyadmin-1  Removed
 ✔Container compose-demo-database-1    Removed
 ✔Network compose-demo_default         Removed
```

docker compose config

```
C:\Users\15L\compose-demo>docker compose config
name: compose-demo
services:
  database:
    environment:
      MYSQL_ROOT_PASSWORD: root123
    image: mysql:8.0
    networks:
      default: null
  phpmyadmin:
    environment:
      PMA_HOST: database
    image: phpmyadmin
    networks:
      default: null
    ports:
      - mode: ingress
        target: 80
        published: "8082"
        protocol: tcp
  redis:
    image: redis
    networks:
      default: null
  web:
    image: nginx
    networks:
      default: null
    ports:
      - mode: ingress
        target: 80
        published: "8888"
        protocol: tcp
networks:
  default:
    name: compose-demo_default
```

Conclusion:

In this experiment, we learned how to use Docker Compose to create and manage a multi-container application using a single docker-compose.yml file. We configured and deployed Nginx and MySQL containers, understood the use of services, images, environment variables, port mapping, and Docker volumes. We also learned to use essential Docker Compose commands to start, stop, restart, and remove containers, and gained practical experience in resolving common issues such as port conflicts. Overall, Docker Compose made managing multiple containers simple, efficient, and organized.